BASIC TERMINOLOGIES AND BASICS OF LINUX

BASIC TERMINOLOGIES

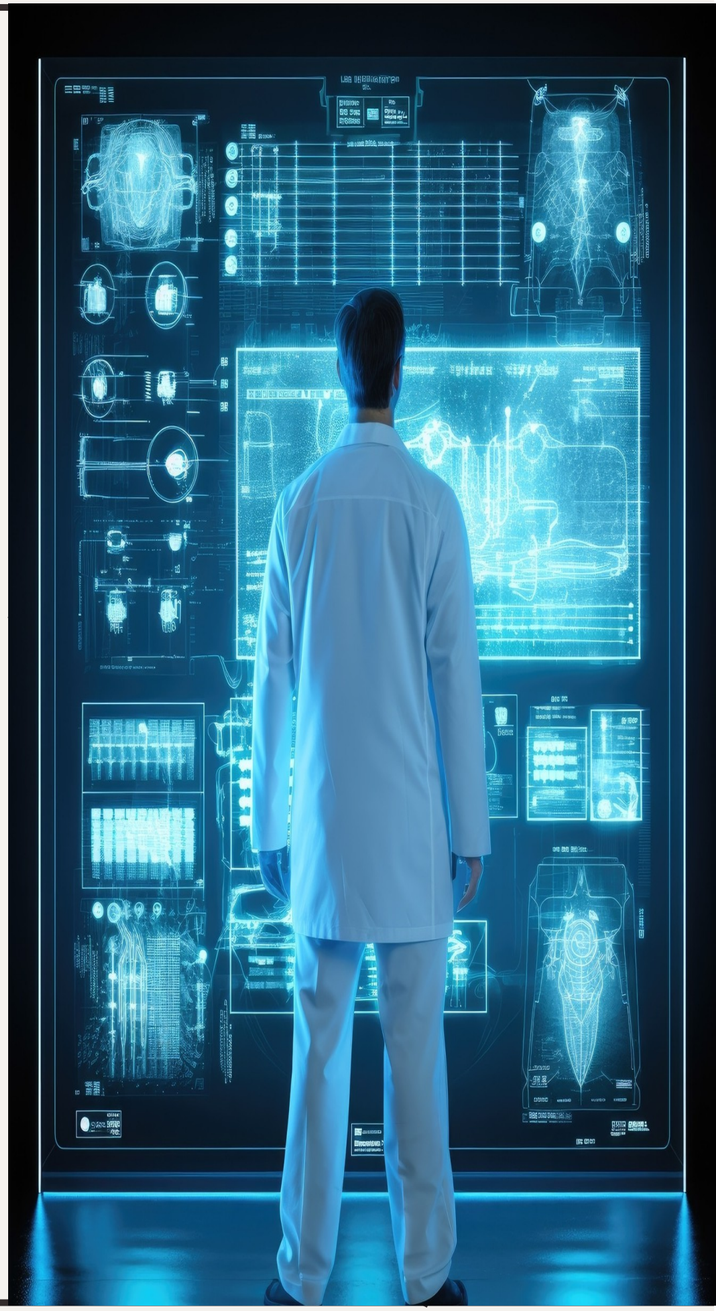
1:BLACK HAT
2:WHITE HAT
3:GRAY HAT
4:RED TEAMING
5:BLUE TEAMING
6:MALWARE
7:TROJAN HORSE
8:WORMS
9:VIRUSES
10:RANSOMWARE
11:ADWARE
12:IP ADDRESS
13:PORT
14:PENETRATION TEST
15:SOCIAL ENGINEERING

https://www.simplilearn.com/top-cybersecurity-terms-you-need-to-know-article

BASIC COMMANDS OF LINUX

COMMANDS	FUNCTIONALITY
whoami	Find out what user we're currently logged in as!
ls	listing
cd	change directory
cat	Concatenate
pwd	print working directory
echo	Output any text we provide
find	find the path or directory of the file Eg. find -name password.txt find –name *.txt

Symbol / Operator	Description
&	This operator allows you to run commands in the background of your terminal.
&&	This operator allows you to combine multiple commands together in one line of your terminal.
>	This operator is a redirector - meaning that we can take the output from a command (such as using cat to output a file) and direct it elsewhere.
>>	This operator does the same function of the > operator but appends the output rather than replacing (meaning nothing is overwritten).

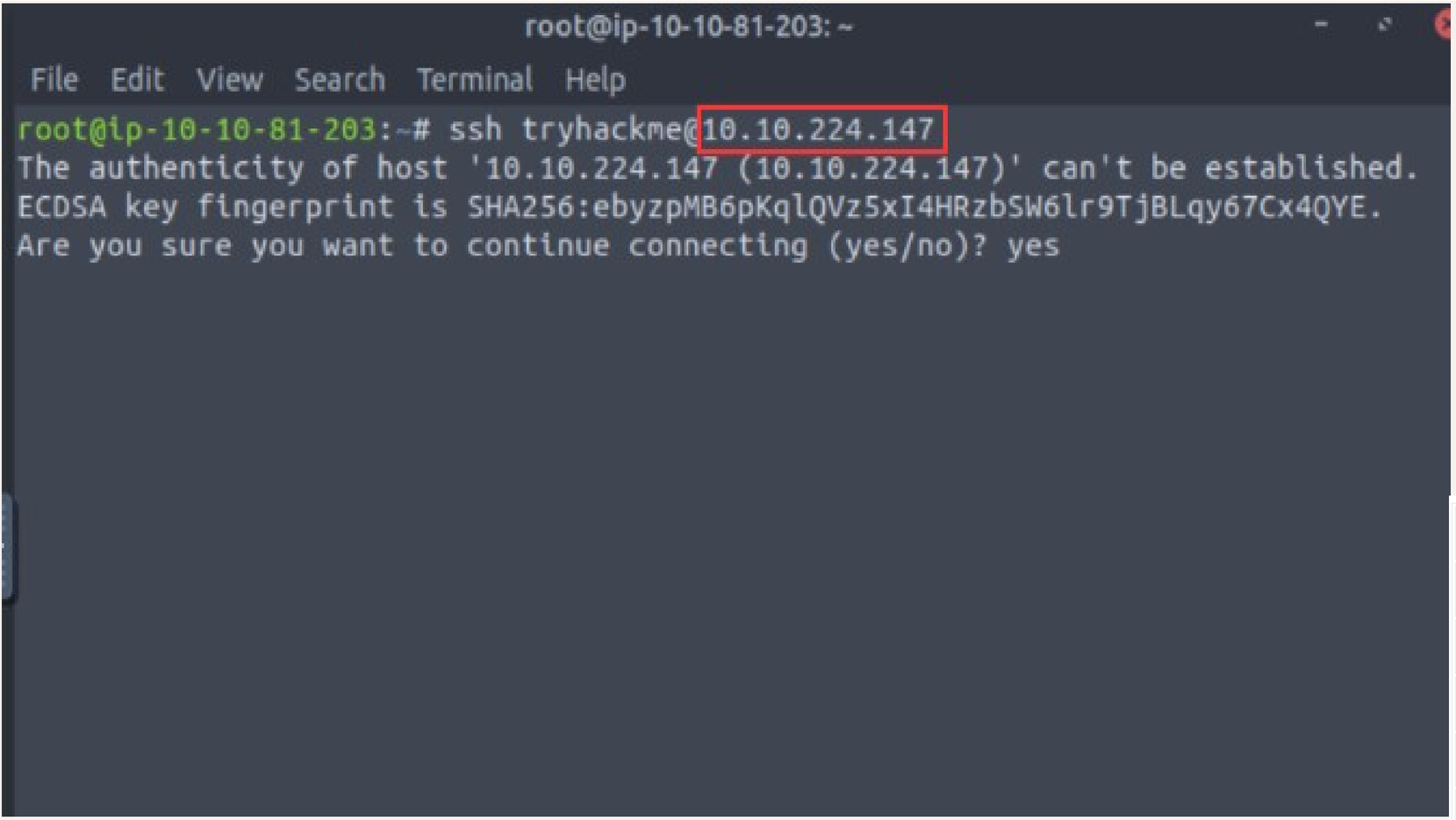


What is SSH & how Does it Work?

Secure Shell or SSH simply is a protocol between devices in an encrypted form. Using cryptography, any input we send in a human-readable format is encrypted for travelling over a network -- where it is then unencrypted once it reaches the remote machine, such as in the diagram below.



- SSH allows us to remotely execute commands on another device remotely.
- Any data sent between the devices is encrypted when it is sent over a network such as the Internet



COMMAND THAT WILL HELP YOU IF YOU DON'T KNOW ITS SYNTAX

1: --help

2: man

Eg. ls –help

man ls

COMMAND THAT OPERATE FILES AND DIRECTORIES

Command	Purpose
touch	Create file
mkdir	Create a folder
cp	Copy a file or folder
mv	Move a file or folder
rm	Remove a file or folder
file	Determine the type of a file

https://tryhackme.com/r/room/linuxfundamentalspart2

ls –ll allows us to see the permission a user has for the files

```
┌──(kali㉿kali)-[~]
└─$ ls -ll
total 108704
-rw-r--r--  1 kali kali       277 Oct 17  2023  123.c
-rw-r--r--  1 kali kali       147 Sep 11  2023  12.c
-rw-r--r--  1 kali kali      1172 Oct 16  2023  1.c
-rw-r--r--  1 kali kali       583 Oct 16  2023  2.c
-rw-r--r--  1 kali kali        73 Oct  3  2023  a1.c
-rw-r--r--  1 kali kali       195 Dec 19 04:23  a.c
-rw-r--r--  1 kali kali        81 Aug 23  2023  af.py
drwxr-xr-x  4 kali kali      4096 Nov 10 12:37  AI
-rwxr-xr-x  1 kali kali     15952 Dec 19 04:23  a.out
-rw-r--r--  1 kali kali        46 Feb  5 05:12  a.py
drwxr-xr-x  3 kali kali      4096 Oct 10  2023  Arduino
-rw-r--r--  1 kali kali       106 Aug 24  2023  asd.c
-rw-------  1 kali kali       176 Jan  7 03:40  a.txt.save
-rw-r--r--  1 kali kali       281 Oct 31  2023  bot.py
drwxr-xr-x  4 kali kali      4096 Dec 12 10:02  Desktop
drwxr-xr-x  4 kali kali      4096 Jan 22 02:14  Documents
drwxr-xr-x  9 kali kali      4096 Mar 19 00:19  Downloads
-rw-r--r--  1 kali kali       364 Dec 19 04:24  example.py
-rw-------  1 kali kali        64 Dec 13 12:38  example.py.save
```



/etc

This root directory is one of the most important root directories on your system. The etc folder (short for etcetera) is a commonplace location to store system files that are used by your operating system.

For example, the sudoers file highlighted in the screenshot below contains a list of the users & groups that have permission to run sudo or a set of commands as the root user.

Also highlighted below are the "**passwd**" and "**shadow**" files. These two files are special for Linux as they show how your system stores the passwords for each user in encrypted formatting called sha512.

Some notable contents of the /etc directory

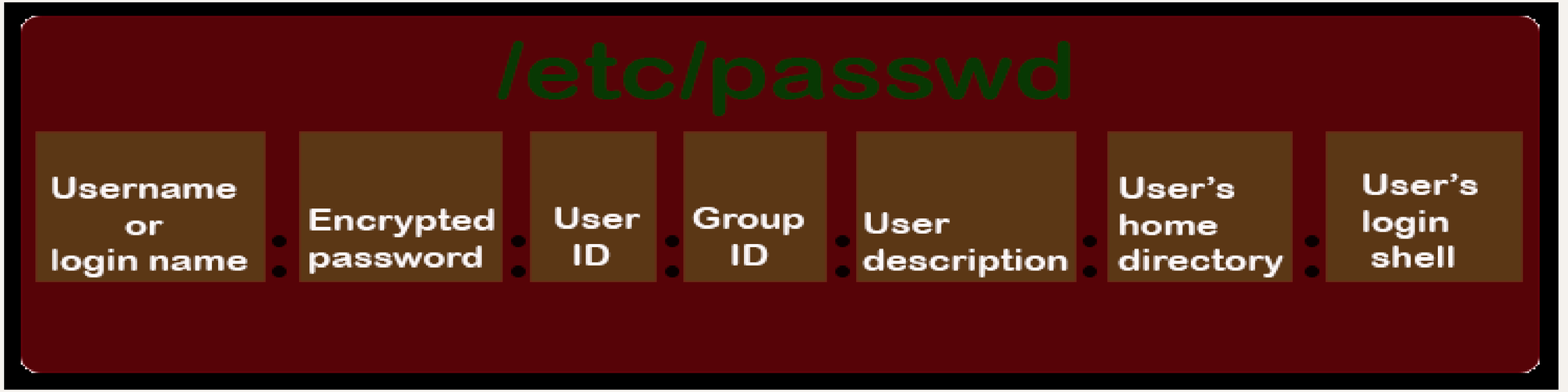
```
tryhackme@linux2:/etc$ ls
shadow passwd sudoers sudoers.d
```

/var

The "/var" directory, with "var" being short for variable data, is one of the main root folders found on a Linux install. This folder stores data that is frequently accessed or written by services or applications running on the system. For example, log files from running services and applications are written here (**/var/log**), or other data that is not necessarily associated with a specific user (i.e., databases and the like).

Some notable contents of the /var directory

```
tryhackme@linux2:/var$ ls
backups log opt tmp
```



/tmp

This is a unique root directory found on a Linux install. Short for "**temporary**", the /tmp directory is volatile and is used to store data that is only needed to be accessed once or twice. Similar to the memory on your computer, once the computer is restarted, the contents of this folder are cleared out.

What's useful for us in pentesting is that any user can write to this folder by default. Meaning once we have access to a machine, it serves as a good place to store things like our enumeration scripts.

Some notable contents of the /tmp directory

```
root@linux2:/tmp# ls
todelete trash.txt rubbish.bin
```

/root

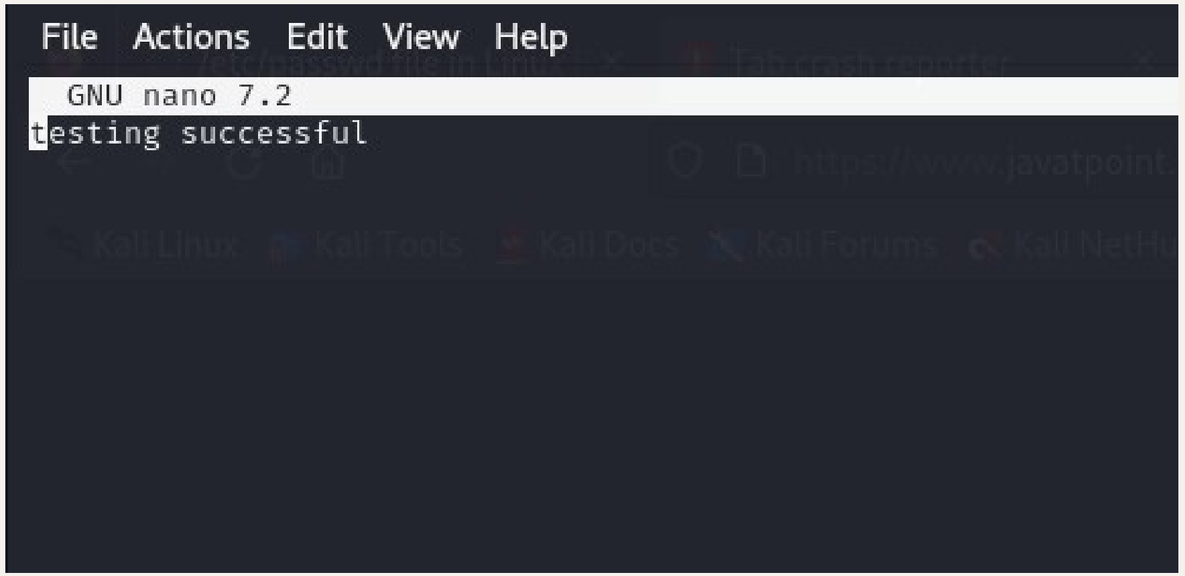
Unlike the **/home** directory, the **/root** folder is actually the home for the "root" system user. There isn't anything more to this folder other than just understanding that this is the home directory for the "root" user. But, it is worth a mention as the logical presumption is that this user would have their data in a directory such as "**/home/root**" by default.

Some notable contents of the /root directory

```
root@linux2:~# ls
myfile myfolder passwords.xlsx
```

FILE EDITORS





LIKE NANO THERE ARE MORE TEXT EDITORS:

1.Vim

2.Gedit

3:Kate

4:Sublime

General Utilities

Downloading Files (Wget)

A pretty fundamental feature of computing is the ability to transfer files. For example, you may want to download a program, a script, or even a picture. Thankfully for us, there are multiple ways in which we can retrieve these files.

We're going to cover the use of `wget`. This command allows us to download files from the web via HTTP -- as if you were accessing the file in your browser. We simply need to provide the address of the resource that we wish to download. For example, if I wanted to download a file named "myfile.txt" onto my machine, assuming I knew the web address it -- it would look something like this:

```
wget https://assets.tryhackme.com/additional/linux-fundamentals/part3/myfile.txt
```

Git clone is used to clone a git repository into our linux machine



THANK YOU

MADE BY ARYAN VERMA